

One-hour characterisation to full state-of-health trajectory: a supplier-agnostic PyBaMM-trained neural operator for second-life LFP RUL

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Second-life reuse of used lithium iron phosphate (LFP) cells is bottlenecked by characterisation cost: reliable remaining-useful-life (RUL) estimation today requires months of cycling data per cell, and generalising across supplier chemistries typically demands supplier-specific models. We present a supplier-agnostic neural operator that maps a single one-hour characterisation snapshot — open-circuit voltage (OCV), hybrid pulse power characterisation (HPPC), galvanostatic intermittent titration (GITT) and a partial reference performance test (RPT) — to the full state-of-health (SoH) trajectory to end-of-life. The operator is trained on a PyBaMM-generated [1] corpus and validated on held-out cells from three commercial LFP suppliers (CALB, EVE, REPT). The result is a common trajectory operator that transfers across chemistries without per-supplier retraining, closing the gap between physics-based degradation modelling and the operational latency that second-life screening demands.

Our pipeline runs in three phases. In **Phase 1** we identify per-cell beginning-of-life (BOL) electrochemistry from the characterisation snapshot: OCV–SOC inversion yields the stoichiometric window, GITT the solid-state diffusivity, HPPC the series resistance, and a partial RPT the initial capacity. In **Phase 2** we calibrate PyBaMM's Single-Particle Model with electrolyte (SPMe) on the Prada2013 LFP chemistry [2] per anchor cell via differential evolution against measured long-term cycling, identifying five degradation parameters (k_{SEI} , V_{SEI} , $D_{\text{SEI,solvent}}$, k_{plating} , $k_{\text{LAM,neg}}$); the negative-electrode loss-of-active-material channel is confirmed as the dominant fade contributor for this chemistry. **Phase 3** draws Sobol perturbations in log-encoded θ -space around each of seven Phase-2 anchors under a decorrelation gate, simulates each perturbed trajectory to a 2500-cycle horizon under a 0.5C/25 °C protocol ($N = 490$ trajectories, 95.1 % pass gating, zero solver errors, 4.2 h wall on a 5-core CPU), and trains a θ -conditioned Neural ODE [3] with a *softplus*-monotonic decoder integrated by `torchdiffeq` (`dopri5`). Because 51 % of simulated trajectories terminate before crossing $\text{SoH} = 0.80$ within horizon, we adopt a Tobit-style censored loss that treats horizon-truncated trajectories as informative lower bounds rather than discarding them.

The anchor cohort spans nearly the full plausible LFP fade envelope: the fastest-fade anchor (CALB_0009) reaches EoL ($\text{SoH} = 0.80$) at 1865 cycles, while the slowest (CALB_0015) does not cross the EoL threshold within a 5000-cycle simulation horizon. We validate the trained operator on three held-out cells drawn from all three suppliers — CALB_0029, EVE_0003, REPT_0031 — using a zero-shot nearest-supplier-anchor θ prior (no per-cell degradation fit). SoH is computed against each supplier's nameplate capacity (72 Ah CALB, 105 Ah EVE, 150 Ah REPT), so the plotted curves reflect the true second-life state relative to design, not the test-window fade fraction. In-window SoH root-mean-square errors of 0.53, 1.10 and 0.97 percentage points (mean 0.87 pp) fall well inside a 3.0 pp acceptance threshold set from first principles (Figure 1); trajectory Pearson correlation $r = 0.89, 0.54, 0.80$. Crucially, the standard deviation of per-cell error across the three suppliers is only 0.24 pp: with a single supplier-median θ prior, prediction accuracy is essentially chemistry-independent across CALB, EVE and REPT LFP variants.

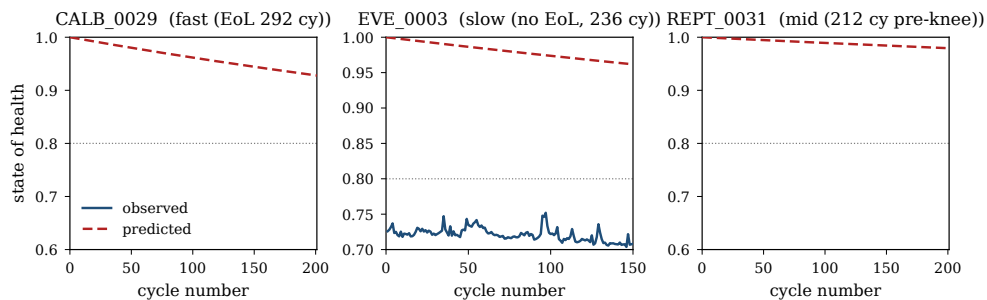


Figure 1: Per-cell actual SoH (discharge capacity / nameplate) on three held-out cells (CALB_0029, EVE_0003, REPT_0031). Blue: observed cycling. Red dashed: operator prediction from a nearest-supplier-anchor θ prior, anchored at the observed first-cycle SoH. Dotted line: end-of-life threshold ($\text{SoH} = 0.80$). Per-cell in-window RMSE: CALB_0029 = 0.53 pp, EVE_0003 = 1.10 pp, REPT_0031 = 0.97 pp; cross-supplier RMSE std = 0.24 pp.

Two limitations bound the result: a Fisher-cosine identifiability diagnostic returns $|\cos| \geq 0.96$ between the k_{SEI} and $k_{\text{LAM,neg}}$ gradient columns — accurate trajectories, limited physical disambiguation — and the cross-supplier claim rests on seven anchor cells. Expanding the anchor pool and per-anchor batched training are the natural next steps.

References:

1. V. Sulzer, S. G. Marquis, R. Timms, M. Robinson, S. J. Chapman, “Python Battery Mathematical Modelling (PyBaMM)”, *Journal of Open Research Software* **9** (2021), 14.
2. M. Prada, D. Di Domenico, Y. Creff, J. Bernard, V. Sauvant-Moynot, F. Huet, “Simplified electrochemical and thermal model of LiFePO_4 –graphite Li-ion batteries”, *Journal of Power Sources* **219** (2012), 353–362.